

The product-vector state on the completed CAR algebra

`MathlibAnnex.CStarAlgebra.CAR.rootState`

def

Compatible evaluation at the distinguished matrix coordinate extends continuously to the CAR completion.

Statement

Let A be the completed CAR algebra: the norm completion of the increasing union of $A_n = M_{2^n}(\mathbb{C})$ under the unital embeddings $a \mapsto a \otimes I_2$ (with the fixed coordinate reindexing). Write $\iota_n : A_n \rightarrow A$ for the canonical isometric inclusion and $e_{ij}^{(n)} = \iota_n(E_{ij})$ for its matrix units. At each stage take $a \mapsto a_{00}$, the vector functional of the distinguished coordinate vector. These functionals determine a continuous linear functional $\omega : A \rightarrow \mathbb{C}$ satisfying $\omega(\iota_n(a)) = a_{00}$ for every n and $a \in A_n$. The definition called the root state is this extension.

Definition

The coordinate estimate $|a_{00}| \leq \|a\|$ bounds each stage functional by 1. The standard inclusion preserves the zero coordinate, so these maps agree on common representatives in the algebraic direct limit. They therefore define a bounded linear functional on the normed direct limit. Extending that functional to its completion gives ω ; the extension keeps the stage values unchanged. Positivity passes from the stage functionals to the completion by continuity, and evaluation at the common unit gives normalization.

Assumptions

The stage system and distinguished zero coordinate are fixed as above. The completion uses the CAR norm. No ambient Hilbert-space representation is chosen in this definition.

Conclusion

The defining restriction formula determines ω uniquely among continuous linear functionals, because the stage union is dense. The cited positivity and normalization results give $\omega(a^*a) \geq 0$ and $\omega(1) = 1$, so this functional is a state.

Normalization and positivity are properties proved after the definition. The state evaluates one distinguished coordinate; it is not the normalized trace on a matrix stage. Its definition does not assert that it is faithful.

Main citations

- Definition and its exact construction — `MathlibAnnex.CStarAlgebra.CAR.rootState`
- Stage coordinate functional — `MathlibAnnex.CStarAlgebra.CAR.rootFunctional`
- Compatibility of the coordinate functional — `MathlibAnnex.CStarAlgebra.CAR.rootFunctional_step`
- Bounded functional before completion — `MathlibAnnex.CStarAlgebra.CAR.preRootFunctional`

- Stage restriction of the extended functional — `MathlibAnnex.CStarAlgebra.CAR.rootState_stage`
- Normalization of the root state — `MathlibAnnex.CStarAlgebra.CAR.rootState_one`
- Positivity of the root state — `MathlibAnnex.CStarAlgebra.CAR.rootState_star_mul_self_nonneg`
- Uniqueness from the dense stage restrictions —
`MathlibAnnex.CStarAlgebra.CAR.eq_rootState_of_restrict`

Lean source signature (exact)

```
noncomputable def rootState : Limit →L[ $\mathbb{C}$ ]  $\mathbb{C}$  :=
  preRootFunctional.extend (UniformSpace.Completion.toComplL : PreCAR →L[ $\mathbb{C}$ ] Limit)
```

In the source	Mathematical meaning
<code>Limit; PreCAR</code>	Respectively A and its dense normed stage union A_{alg} .
<code>rootState : Limit →L[$\mathbb{C}$] $\mathbb{C}$</code>	The continuous complex-linear functional $\omega : A \rightarrow \mathbb{C}$.
<code>preRootFunctional</code>	On A_{alg} , the bounded functional with value a_{00} on a stage matrix a .
<code>UniformSpace.Completion.toComplL : PreCAR →L[$\mathbb{C}$] Limit</code>	The canonical inclusion $A_{\text{alg}} \rightarrow A$ of the dense union into its completion.
<code>preRootFunctional.extend (...)</code>	Extend that same coordinate functional continuously along this inclusion, obtaining ω with $\omega(\iota_n(a)) = a_{00}$. The definition chooses no ambient representation; statehood and purity are later properties.

Further source notes: The stage inclusion ι_n used in the Statement is named `ofStage n` in the separately linked restriction theorem `rootState_stage`, not in the defining line displayed here. Positivity and normalization are also separate cited results.

Earlier published Card and PDF

▼ **Exact content identity**

Declaration: `MathlibAnnex.CStarAlgebra.CAR.rootState`
 Accepted content SHA-256: 1704b60839cec4bda29d82b73690aef3c3ee6e19622973f31b2c0358d9c52fab
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